String, StringBuffer, StringBuilder, Math and Wrapper Classes

String Class

1. **String Operations**
   - Write a program to take a sentence as input and:
     - Find its length.
     - Extract a substring from index 5 to 10.
     - Convert the entire string to uppercase and lowercase.
2. **Palindrome Check**
   - Create a program to check whether a given string is a palindrome.
3. **Anagram Checker**
   - Write a program to check if two given strings are anagrams of each other.
4. **Character Frequency**
   - Create a program to count the occurrence of each character in a string.
5. **Word Reverse**
   - Write a program to reverse the words in a given sentence (e.g., "Hello World" → "olleH dlroW").

StringBuffer Class

6. **Mutable String Operations**
   - Use StringBuffer to perform the following on a string:
     - Append " Java Programming" to the string.
     - Insert "is fun " at the 6th position.
     - Replace "Programming" with "Language".
     - Reverse the entire string.
7. **Performance Comparison**
   - Compare the performance of String concatenation and StringBuffer when appending 1000 characters in a loop.
8. **Vowel Remover**
   - Write a program that removes all vowels from a given string using StringBuffer.

StringBuilder Class

9. **Efficient String Modifications**
   - Write a program using StringBuilder to perform the following on a string:
     - Delete characters between index 2 and 5.

        - Append " Efficient Programming".
        - Reverse the entire string.
10. **Word Count and Modification**
    - Use StringBuilder to count the number of words in a sentence and append the word count at the end.
11. **Performance Benchmarking**
    - Write a program to compare the performance of String, StringBuffer, and StringBuilder for concatenating 10,000 strings.

Math Class

12. **Math Operations**
    - Write a program to take two numbers as input and perform the following:
        - Calculate their power (x^y).
        - Find the square root of both numbers.
        - Calculate the absolute difference between them.
13. **Random Number Generation**
    - Generate 10 random integers between 1 and 100 using the Math.random() method.
14. **Circle Calculations**
    - Create a program to calculate the area and circumference of a circle using Math.PI and Math.pow().
15. **Number Rounding**
    - Write a program to demonstrate the use of Math.ceil(), Math.floor(), and Math.round() with a floating-point number.
16. **Trigonometric Functions**
    - Write a program to calculate the sine, cosine, and tangent values for angles between 0° and 360°.

Wrapper Classes

17. **Boxing and Unboxing**
    - Write a program to demonstrate:
        - Boxing a primitive int into an Integer object.
        - Unboxing the Integer object back to an int.
18. **Auto-boxing and Auto-unboxing**
    - Create a program to add primitive int values to a list of Integer type using auto-boxing, and calculate the sum using auto-unboxing.
19. **Data Conversion**
    - Write a program that takes a string input of a number and converts it into:
        - int using Integer.parseInt().
        - double using Double.parseDouble().
        - long using Long.parseLong().
20. **Primitive-to-Wrapper Comparison**

- Create a program to compare primitive `int` and wrapper `Integer` values using `.equals()` and `==` operators.